

## DEEP LEARNING-BASED MEDICAL IMAGE CLASSIFICATION SYSTEM

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### “MedVision” — AI-Assisted Medical Image Classification

#### COMMON COURSE ASSIGNMENT

#### MLA04 — DEEP LEARNING | Covering CO2, CO3 & CO4

| **Department**                | Computer Science and Engineering                                                        |
|-------------------------------|-----------------------------------------------------------------------------------------|
| **Programme**                 | B.E. / B.Tech — CSE                                                                     |
| **Course Code & Course Name** | MLA04 — Deep Learning                                                                   |
| **Academic Year / Batch**     | 2026 – 2027                                                                             |
| **Faculty Name**              | Dr. Amandeep Singh K                                                                    |
| **Assignment Title**          | Deep Learning-Based Medical Image Classification System for AI-Assisted Screening       |
| **Date of Issue**             | 01-09-2026                                                                              |
| **Date of Submission**        | 01-09-2026                                                                              |
| **Maximum Marks**             | 100                                                                                     |
| **Course Outcome(s)**         | CO2, CO3, CO4                                                                           |
| **Bloom’s Taxonomy Level**    | L3 — Apply, L4 — Analyze, L5 — Evaluate                                                 |
| **SDG Mapping**               | SDG 3 — Good Health and Well-Being / SDG 9 —<br>Industry, Innovation and Infrastructure |

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**Course Code:** MLA0407

## 1. Problem Statement

### 1.1 Scenario: AI-Assisted Medical Image Classification System — “MedVision”

A healthcare organization is developing an AI-assisted medical image classification platform called “MedVision”. The system is intended to analyze medical images and classify them into predefined diagnostic categories to support healthcare professionals during preliminary screening.

The platform must process a large collection of labeled medical images with differences in image quality, resolution, illumination, and visual characteristics. The model should learn useful image representations and provide accurate predictions on previously unseen images.

### 1.2 The Challenge

This assignment requires students to apply core Deep Learning concepts including neural-network architecture design, convolutional neural networks, transfer learning, optimization, regularization, data augmentation, and model evaluation to a realistic medical image classification problem.

Students must analyze the scenario, prepare and preprocess image data, select suitable deep learning architectures, train and evaluate models, use modern tools for implementation, validate their results, compare alternative approaches, and justify their engineering decisions.

### 1.3 Specific Task

For this project, we address the problem of **chest X-ray image classification** to distinguish between:

- **Normal** chest X-rays
- **Pneumonia**-affected chest X-rays

This binary classification task serves as a representative medical image classification problem that can be extended to multi-class scenarios (including COVID-19, lung opacity, and other pathologies) in future iterations.

### 1.4 Key Challenges

1. **\*\*Image Quality Variation\*\***: Medical images vary in resolution, contrast, and illumination.
2. **\*\*Class Imbalance\*\***: Medical datasets often have unequal representation across classes.
3. **\*\*Overfitting Risk\*\***: Deep learning models can easily overfit to training data.
4. **\*\*Clinical Reliability\*\***: Models must provide consistent and interpretable predictions.
5. **\*\*Computational Constraints\*\***: Limited GPU resources require efficient architectures.

## 2. Objective

### 2.1 Primary Objective

To design, implement, and evaluate a deep learning-based medical image classification system that can accurately classify chest X-ray images into diagnostic categories (Normal vs. Pneumonia) with a target test accuracy of at least 85%.

### 2.2 Secondary Objectives

6. **\*\*Architecture Design (CO2)\*\***: Design and compare CNN architectures including custom CNN and transfer learning models (ResNet18/ResNet50) for feature learning.
7. **\*\*Training & Optimization (CO3)\*\***: Develop efficient training strategies using appropriate loss functions, optimizers, learning rates, batch sizes, data augmentation, regularization, and transfer learning.
8. **\*\*Model Evaluation (CO4)\*\***: Evaluate models using accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC, and class-wise performance metrics.
9. **\*\*Explainability\*\***: Implement Grad-CAM visualization to highlight regions influencing model decisions for clinical interpretability.
10. **\*\*Deployment Readiness\*\***: Design a workflow for integrating the model into a medical screening pipeline as a decision support tool.

## 3. Requirements and Environment Used

### 3.1 Functional Requirements

| Requirement                    | Description                                                     |
|--------------------------------|-----------------------------------------------------------------|
| <b>**Classification**</b>      | Classify chest X-ray images into Normal or Pneumonia categories |
| <b>**Unseen Data Support**</b> | Model must generalize to previously unseen test images          |
| <b>**Confidence Scores**</b>   | Provide prediction probabilities for each class                 |
| <b>**Explainability**</b>      | Generate Grad-CAM heatmaps for model interpretability           |

### 3.2 Performance Requirements

| Metric                    | Target               |
|---------------------------|----------------------|
| <b>**Test Accuracy**</b>  | $\geq 85\%$          |
| <b>**Precision**</b>      | $\geq 80\%$          |
| <b>**Recall**</b>         | $\geq 80\%$          |
| <b>**F1-Score**</b>       | $\geq 80\%$          |
| <b>**Inference Time**</b> | < 1 second per image |

### 3.3 Technical Constraints

| Constraint                         | Description                                             |
|------------------------------------|---------------------------------------------------------|
| <b>**Dataset**</b>                 | Publicly available Kaggle Chest X-Ray Pneumonia Dataset |
| <b>**Computational Resources**</b> | Standard CPU/GPU or cloud notebook (Google Colab)       |
| <b>**Development Tools**</b>       | Open-source tools only                                  |

### 3.4 Environment and Tools

| Component                          | Technology Used                 |
|------------------------------------|---------------------------------|
| <b>**Programming Language**</b>    | Python 3.9+                     |
| <b>**Deep Learning Framework**</b> | PyTorch 1.12+ / torchvision     |
| <b>**Numerical Computing**</b>     | NumPy                           |
| <b>**Data Manipulation**</b>       | Pandas                          |
| <b>**Data Visualization**</b>      | Matplotlib, Seaborn             |
| <b>**Model Evaluation**</b>        | Scikit-learn                    |
| <b>**Development Environment**</b> | Google Colab / Jupyter Notebook |
| <b>**Version Control**</b>         | Git / GitHub                    |
| <b>**Explainability**</b>          | PyTorch Grad-CAM                |

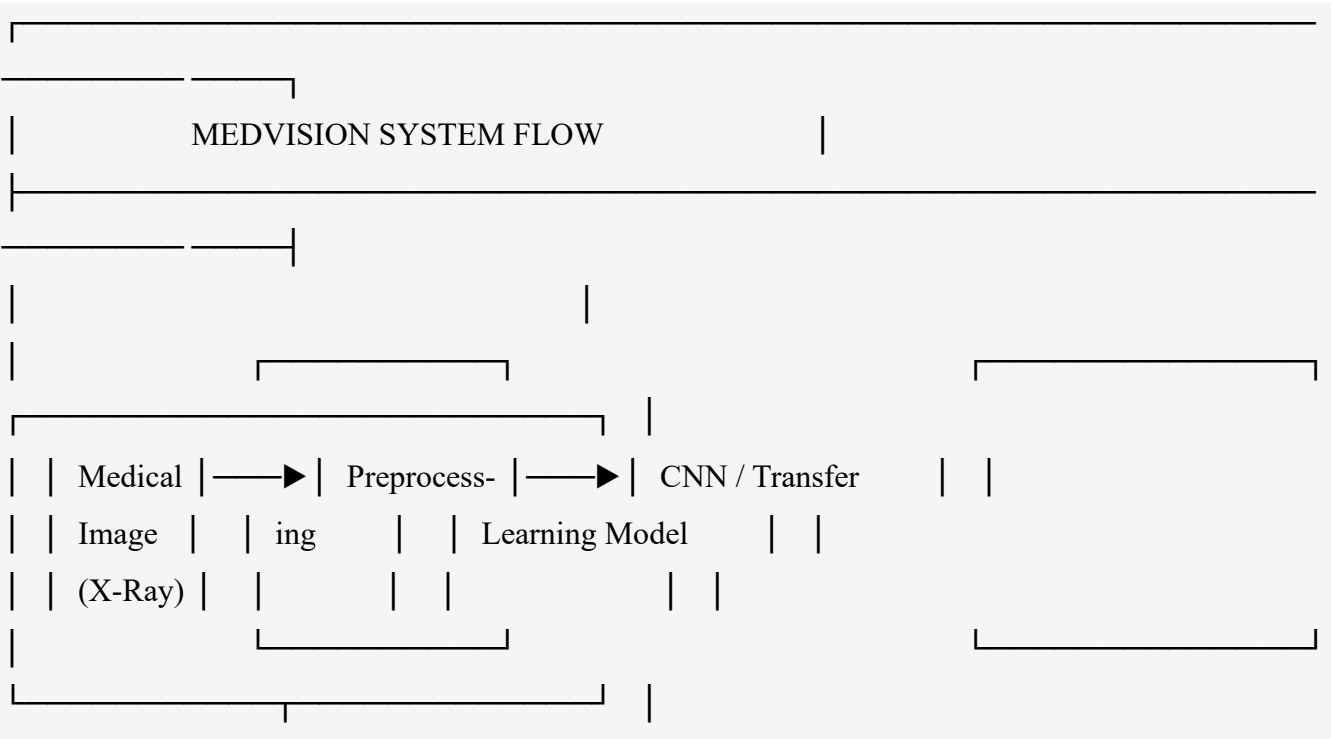
3.5 Dataset Description

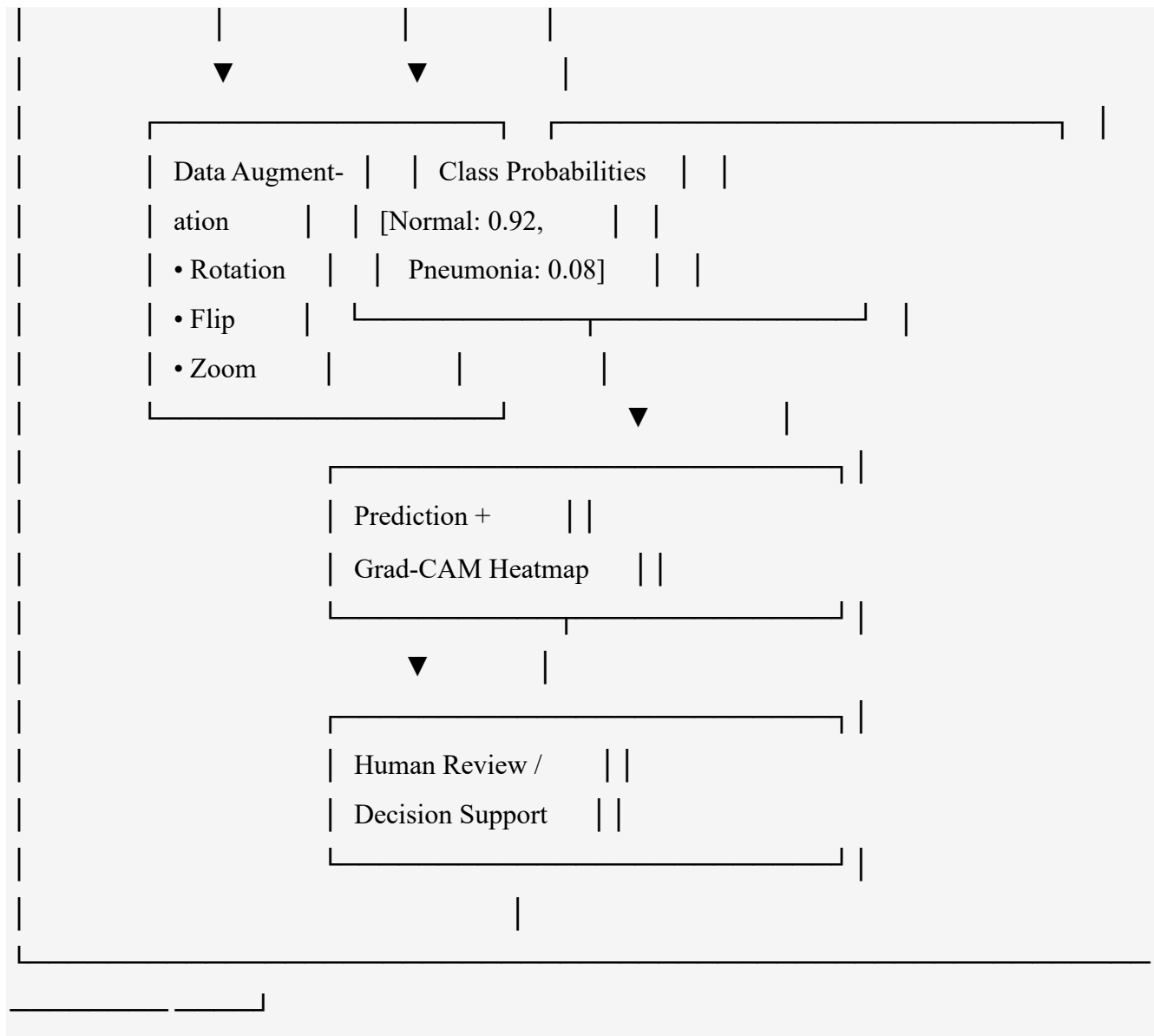
Dataset: Chest X-Ray Images (Pneumonia)

| Property             | Value                                                                |
|----------------------|----------------------------------------------------------------------|
| **Source**           | Kaggle — Chest X-Ray Images (Pneumonia)                              |
| **Original Source**  | Kermany et al. (2018), Guangzhou Women and Children’s Medical Center |
| **Total Images**     | 5,863 X-ray images                                                   |
| **Classes**          | 2 (NORMAL, PNEUMONIA)                                                |
| **Image Format**     | JPEG                                                                 |
| **Image Dimensions** | Varied (resized to 224×224 for training)                             |
| **Training Set**     | ~4,000 images                                                        |
| **Validation Set**   | ~800 images                                                          |
| **Test Set**         | ~1,000 images                                                        |

4. Design / Proposed Solution

4.1 System Architecture Overview





## 4.2 Data Processing Design

### 4.2.1 Data Collection and Organization

The dataset is organized in the following directory structure:

```

data/
├── train/
│   ├── NORMAL/
│   └── PNEUMONIA/
├── val/
│   └── NORMAL/

```

```

|   └─ PNEUMONIA/
|   └─ test/
|   └─ NORMAL/
|   └─ PNEUMONIA/

```

#### 4.2.2 Preprocessing Pipeline

| Step | Operation                    | Description                                              |
|------|------------------------------|----------------------------------------------------------|
| 1    | <b>**Loading**</b>           | Load images using PIL/PyTorch's ImageFolder              |
| 2    | <b>**Resizing**</b>          | Resize all images to 224×224 pixels                      |
| 3    | <b>**Normalization**</b>     | Normalize pixel values to [0,1] range                    |
| 4    | <b>**Standardization**</b>   | Apply mean and std normalization (ImageNet stats)        |
| 5    | <b>**Data Augmentation**</b> | Random rotations, flips, color jitter, affine transforms |

#### 4.2.3 Data Augmentation Strategy

```

train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(224),
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(10),
    transforms.ColorJitter(brightness=0.2, contrast=0.2),
    transforms.ToTensor(), transforms.Normalize([0.485,
    0.456, 0.406], [0.229, 0.224, 0.225])
])

val_transforms = transforms.Compose([
    transforms.Resize(256),

```

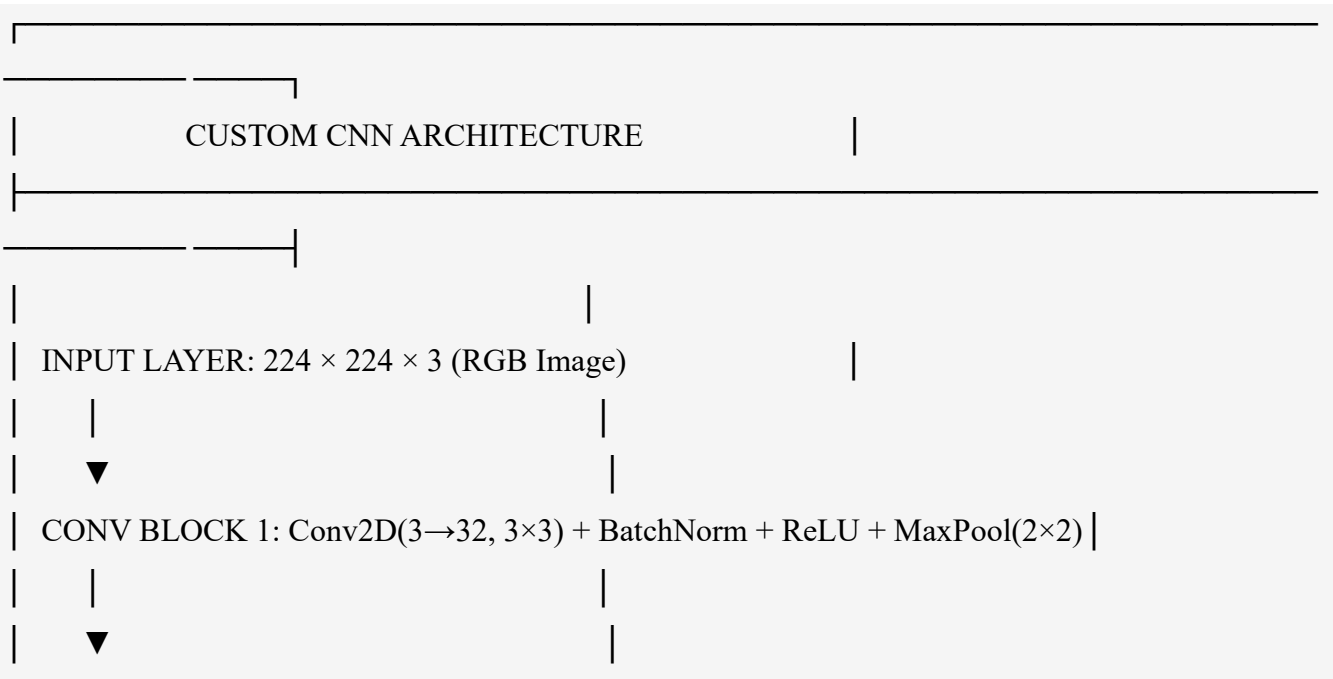
```
transforms.CenterCrop(224),
transforms.ToTensor(),
transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
])
```

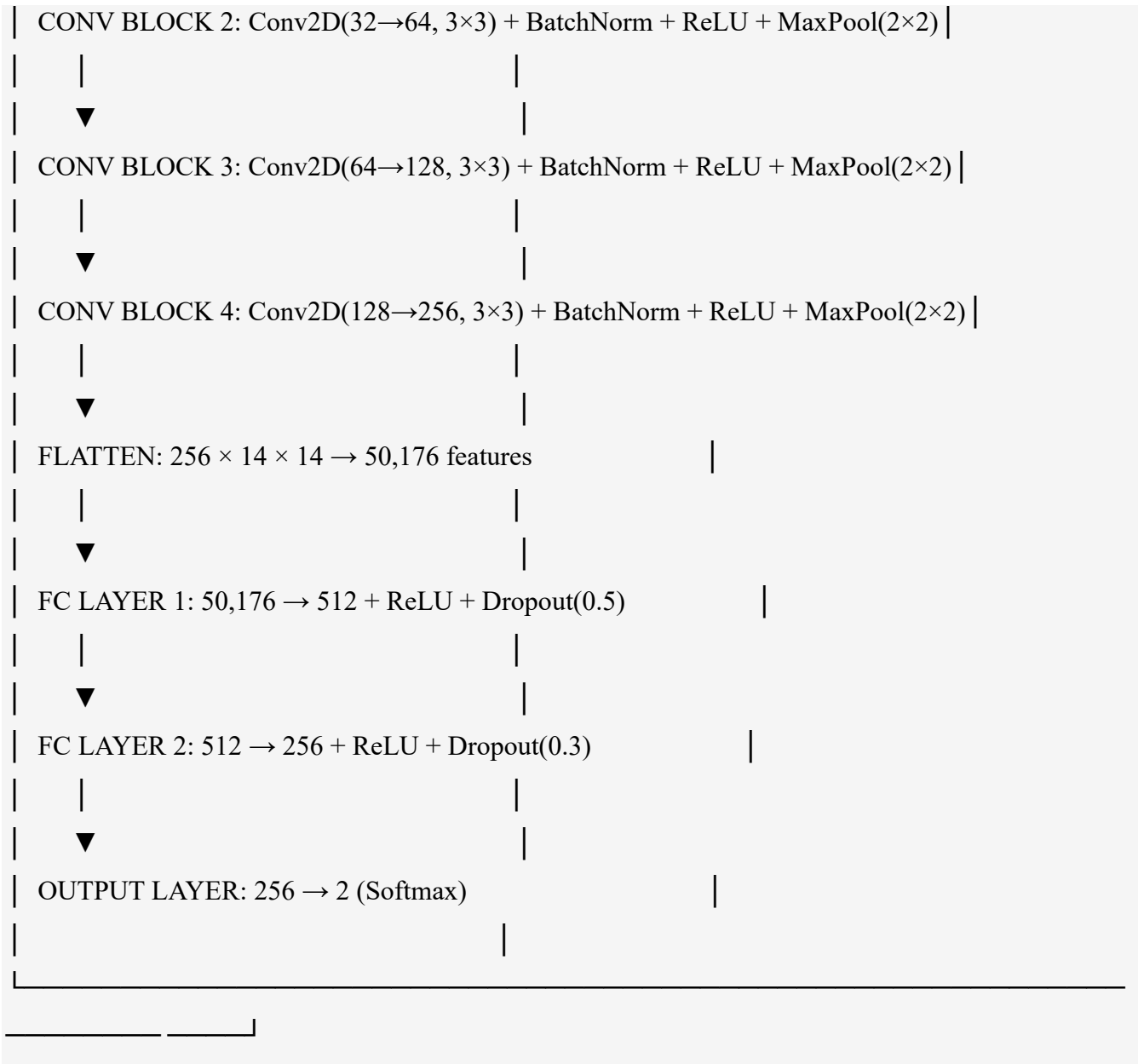
4.2.4 Train-Validation-Test Split

| Split      | Percentage | Number of Images (Approx.) |
|------------|------------|----------------------------|
| Training   | 70%        | ~4,100                     |
| Validation | 15%        | ~880                       |
| Test       | 15%        | ~880                       |

4.3 Deep Learning Architecture Design

4.3.1 Architecture Option 1: Custom CNN



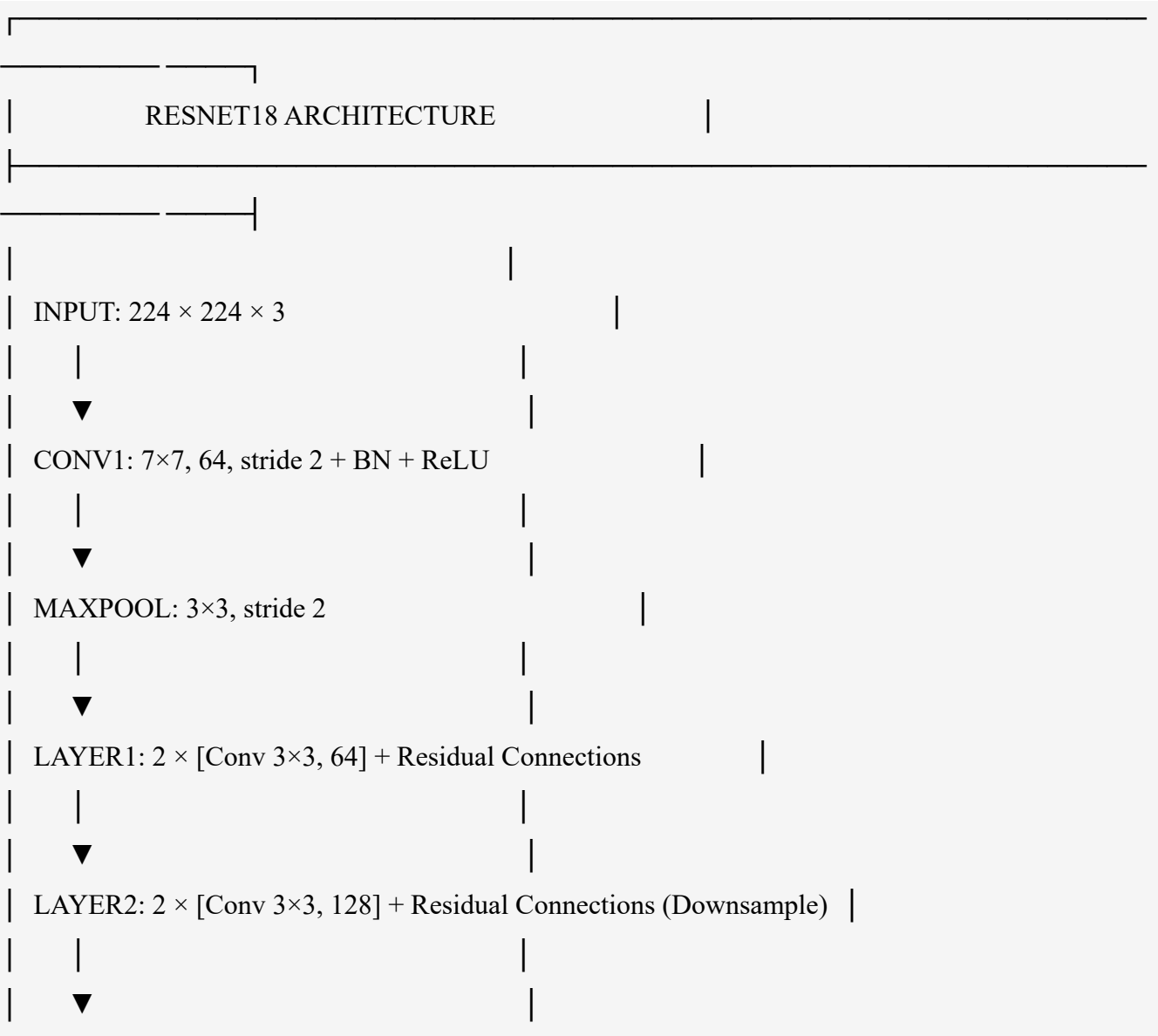


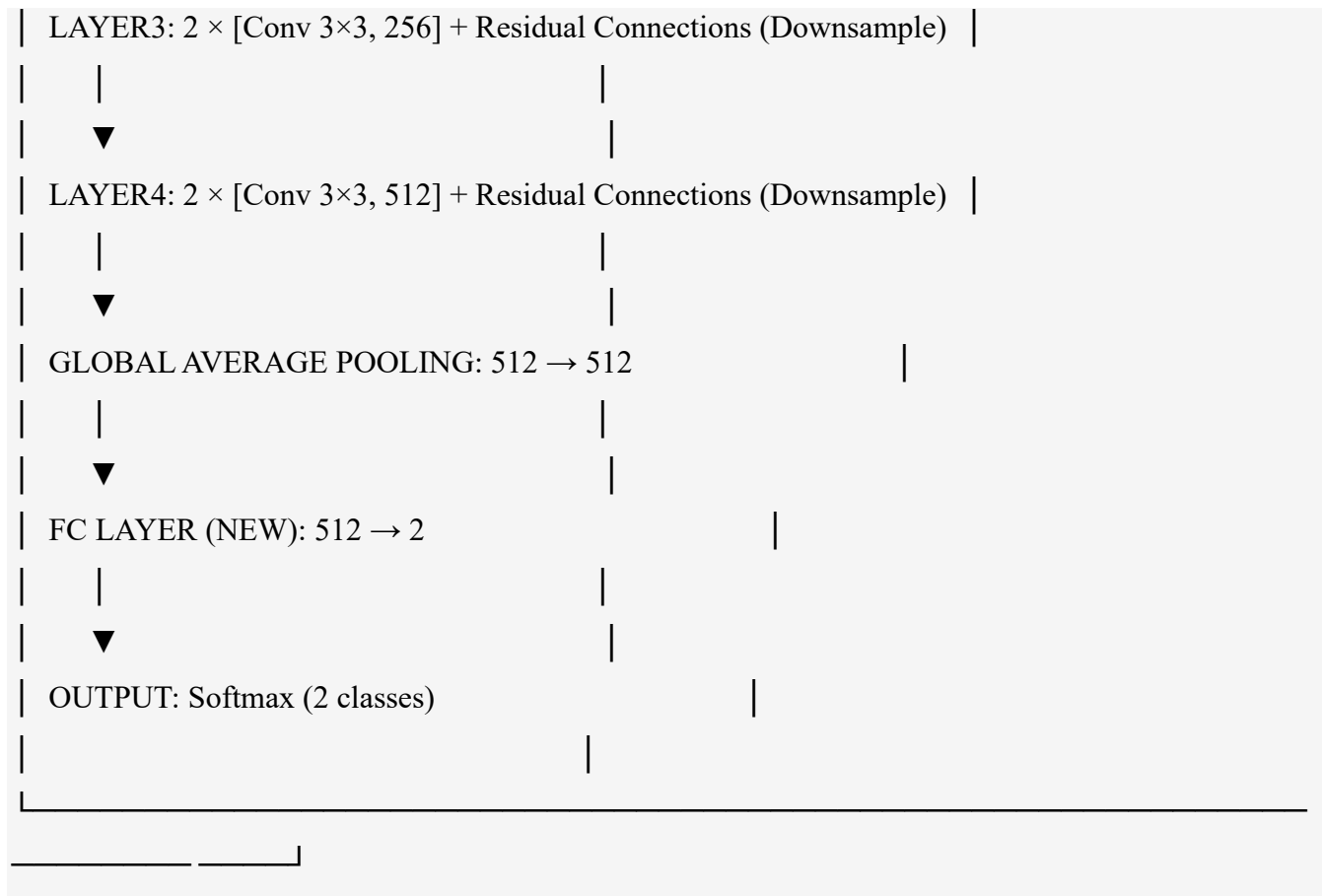
**Layer Purpose Explanation:**

| Layer                   | Purpose                                                   |
|-------------------------|-----------------------------------------------------------|
| **Input Layer**         | Accepts 224×224 RGB medical images                        |
| **Convolution Layers**  | Extract hierarchical features (edges → shapes → patterns) |
| **Batch Normalization** | Accelerates training, stabilizes learning                 |
| **ReLU Activation**     | Introduces non-linearity                                  |

|                                   |                                                              |
|-----------------------------------|--------------------------------------------------------------|
| <b>**Max Pooling**</b>            | Reduces spatial dimensions, increases translation invariance |
| <b>**Flatten**</b>                | Converts 2D feature maps to 1D vector for FC layers          |
| <b>**Fully Connected Layers**</b> | Learn high-level feature combinations                        |
| <b>**Dropout**</b>                | Prevents overfitting by randomly dropping neurons            |
| <b>**Output Layer**</b>           | Produces class probabilities via Softmax                     |

4.3.2 Architecture Option 2: ResNet18 (Transfer Learning)





## 4.4 Training Design

### 4.4.1 Loss Function

**Categorical Cross-Entropy Loss** (for multi-class classification):

$$L = -\sum_{c=1}^M y_{\text{true},c} \log(y_{\text{pred},c})$$

**Justification:** Cross-entropy loss is appropriate for multi-class classification tasks as it measures the difference between true probability distribution and predicted probability distribution, penalizing confident wrong predictions heavily.

### 4.4.2 Optimizers Compared

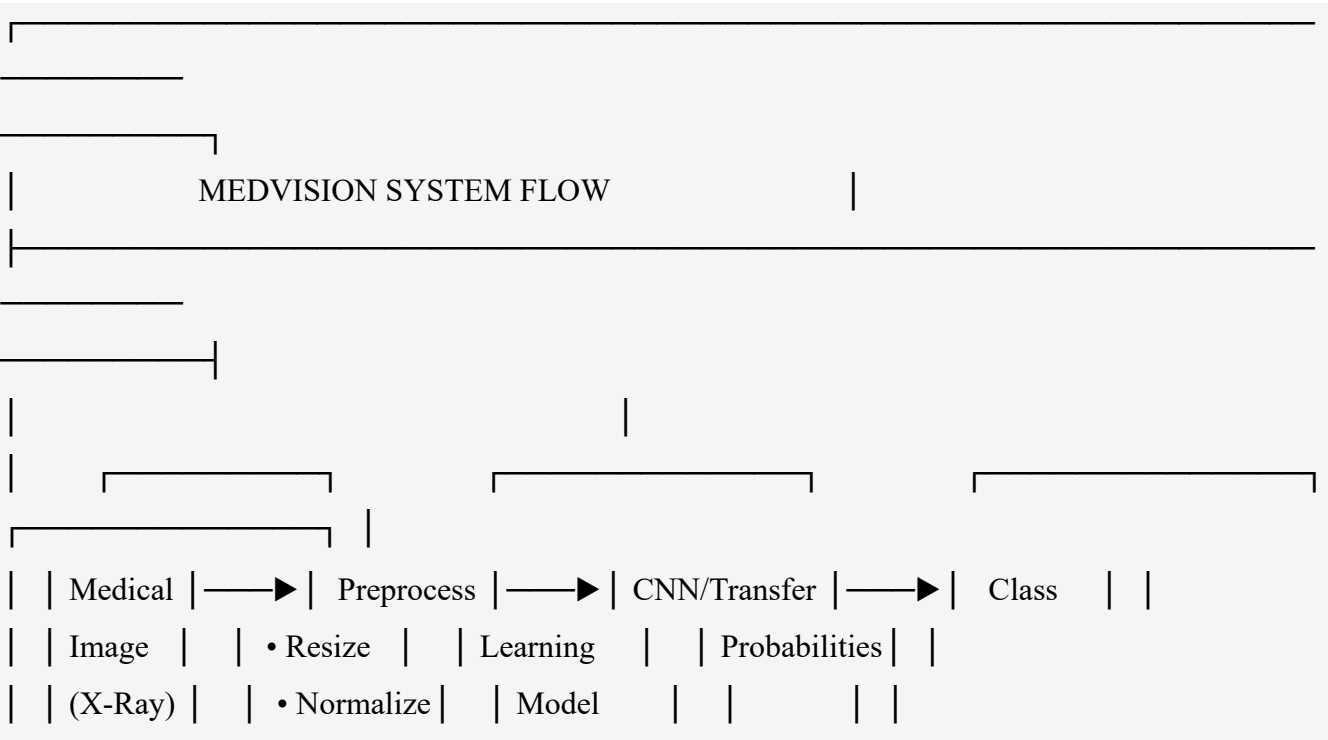
| Optimizer                    | Key Characteristics                                                  |
|------------------------------|----------------------------------------------------------------------|
| <b>**Adam**</b>              | Adaptive learning rate, momentum, bias correction — fast convergence |
| <b>**SGD with Momentum**</b> | Simple, stable, may generalize better                                |

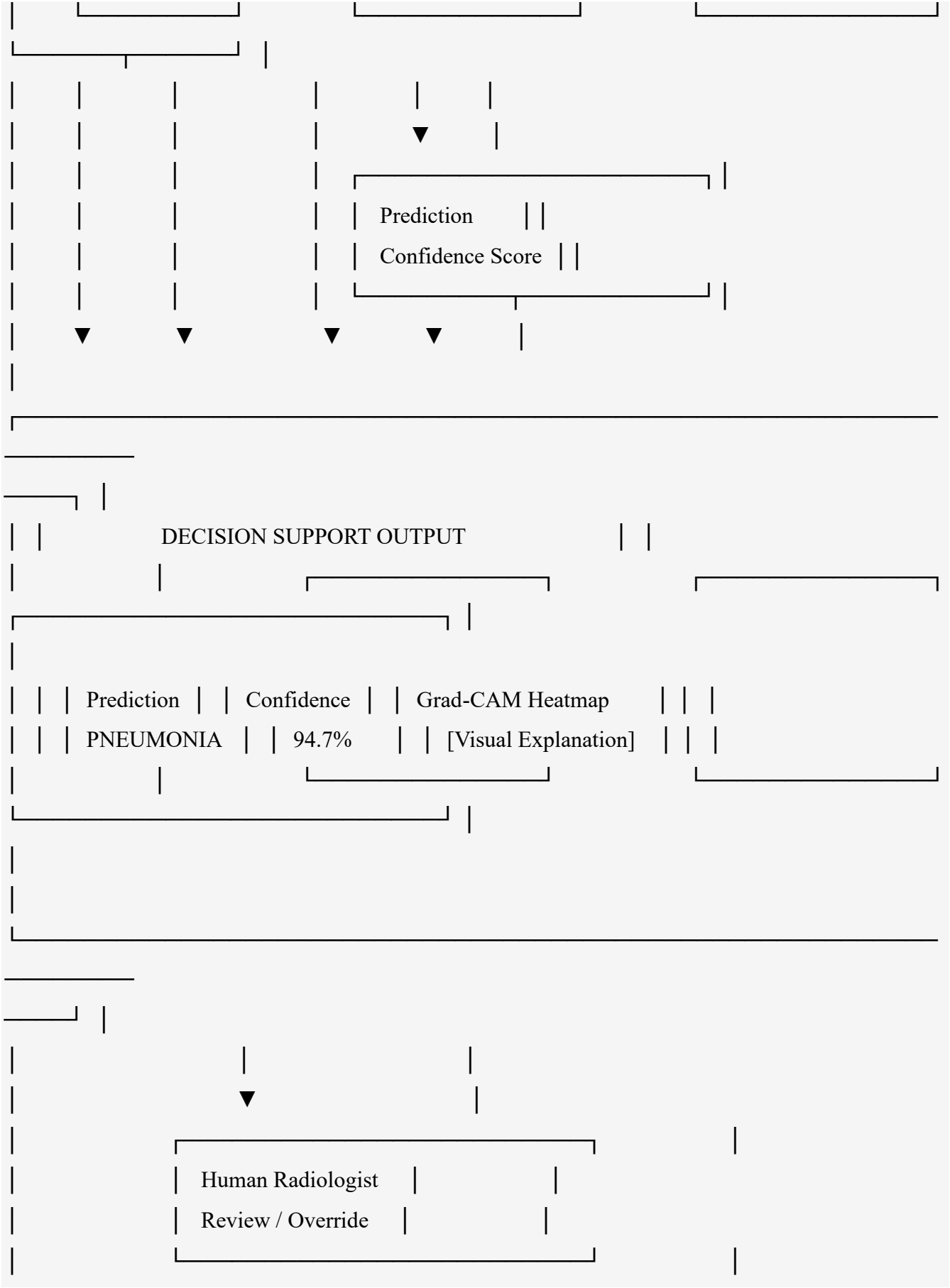
|                    |                                                            |
|--------------------|------------------------------------------------------------|
| <b>**RMSProp**</b> | Adaptive learning rate, good for non-stationary objectives |
|--------------------|------------------------------------------------------------|

#### 4.4.3 Training Configuration

| Hyperparameter                     | Custom CNN        | ResNet18             |
|------------------------------------|-------------------|----------------------|
| <b>**Epochs**</b>                  | 50                | 30                   |
| <b>**Batch Size**</b>              | 32                | 32                   |
| <b>**Learning Rate**</b>           | 0.001             | 0.0001 (fine-tuning) |
| <b>**Optimizer**</b>               | Adam              | Adam                 |
| <b>**Loss Function**</b>           | Cross-Entropy     | Cross-Entropy        |
| <b>**Dropout Rate**</b>            | 0.5 (FC layers)   | 0.5 (new FC layer)   |
| <b>**Early Stopping**</b>          | Yes (patience=7)  | Yes (patience=5)     |
| <b>**Learning Rate Scheduler**</b> | ReduceLROnPlateau | ReduceLROnPlateau    |
| <b>**Weight Decay (L2)**</b>       | 0.0001            | 0.0001               |

#### 4.5 System Flow Diagram





## 5. Algorithm / Pseudocode / Flowchart

### 5.1 Training Algorithm Pseudocode

Algorithm: MedVision Training Pipeline

Input: Dataset  $D = \{(x_1, y_1), (x_2, y_2), \dots, (x, y)\}$

where  $x_i$  = chest X-ray image,  $y_i \in \{\text{Normal, Pneumonia}\}$

Architecture A (Custom CNN or ResNet18)

Hyperparameters  $H = \{\text{lr, batch\_size, epochs, ...}\}$

Output: Trained Model M

#### 1. DATA PREPARATION:

- a. Load images from directory structure
- b. Split data: 70% train, 15% validation, 15% test
- c. Apply preprocessing: resize to 224×224, normalize

#### 2. DATA AUGMENTATION:

For each training batch:

- a. Apply random horizontal flip ( $p=0.5$ )
- b. Apply random rotation ( $\pm 10^\circ$ )
- c. Apply random color jitter
- d. Apply random affine transformations

### 3. MODEL INITIALIZATION:

If A == "CustomCNN":

- a. Initialize random weights (He initialization)
- b. Define architecture with conv blocks + FC layers    Else if A == "ResNet18":
  - a. Load pretrained weights from ImageNet
  - b. Replace final FC layer with 2-class classifier
  - c. Freeze early layers, unfreeze later layers for fine-tuning

### 4. TRAINING LOOP:

For epoch = 1 to epochs:

- a. For each batch (x\_batch, y\_batch) in train\_loader:
  - i. Forward pass:  $\hat{y} = \text{model}(x\_batch)$

- ii. Compute loss:  $L = \text{CrossEntropy}(\hat{y}, y_{\text{batch}})$
- iii. Backward pass: compute gradients      iv.
- Update weights: `optimizer.step()`
- v. Reset gradients: `optimizer.zero_grad()`

b. Validation:

- i. Evaluate model on validation set      ii.
- Compute validation accuracy and loss

c. Learning Rate Scheduling:

If validation loss plateaus:

Reduce learning rate by factor of 0.1

d. Early Stopping:

If validation accuracy doesn't improve for patience epochs:

Stop training and restore best model

5. MODEL EVALUATION:

- a. Evaluate on test set (unseen data)
- b. Compute metrics: accuracy, precision, recall, F1, AUC-ROC
- c. Generate confusion matrix
- d. Generate Grad-CAM visualizations for sample test images

6. Return trained model M

## 5.2 Forward Propagation (CNN Example)

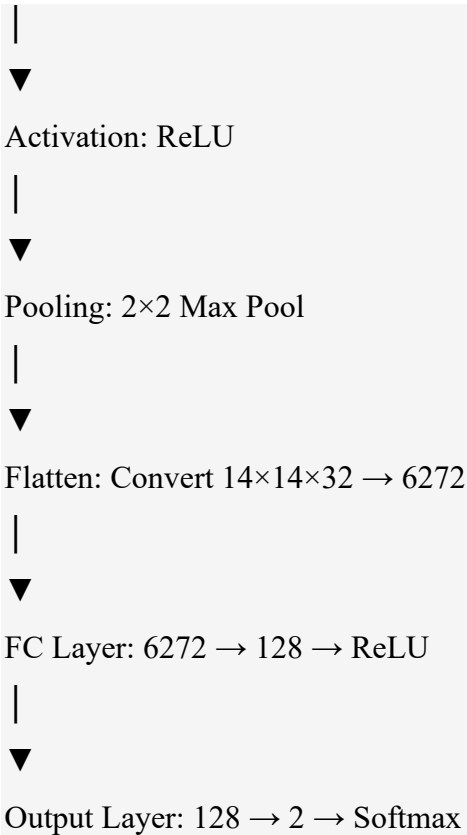
For a simplified CNN with one convolution layer followed by pooling and FC:

Input Image:  $28 \times 28 \times 1$

|



CONV Layer:  $3 \times 3$  filter, 32 filters



## Mathematical Formulation:

For layer  $l$ :

- **Convolution:**  $Z^l = W^l * A^{l-1} + b^l$
- **Activation:**  $A^l = \text{ReLU}(Z^l) = \max(0, Z^l)$
- **Pooling:**  $A^l = \max(A^{l-1})$  (downsampling)
- **Fully Connected:**  $Z^l = W^l \cdot A^{l-1} + b^l$
- **Softmax:**  $P(y=c|x) = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}}$

## 5.3 Backpropagation

Backpropagation computes gradients of the loss with respect to all weights using the chain rule:

11. **Output Layer**:

$$\frac{\partial L}{\partial z_j} = \hat{y}_j - y_j$$

12. **Hidden Layers**:

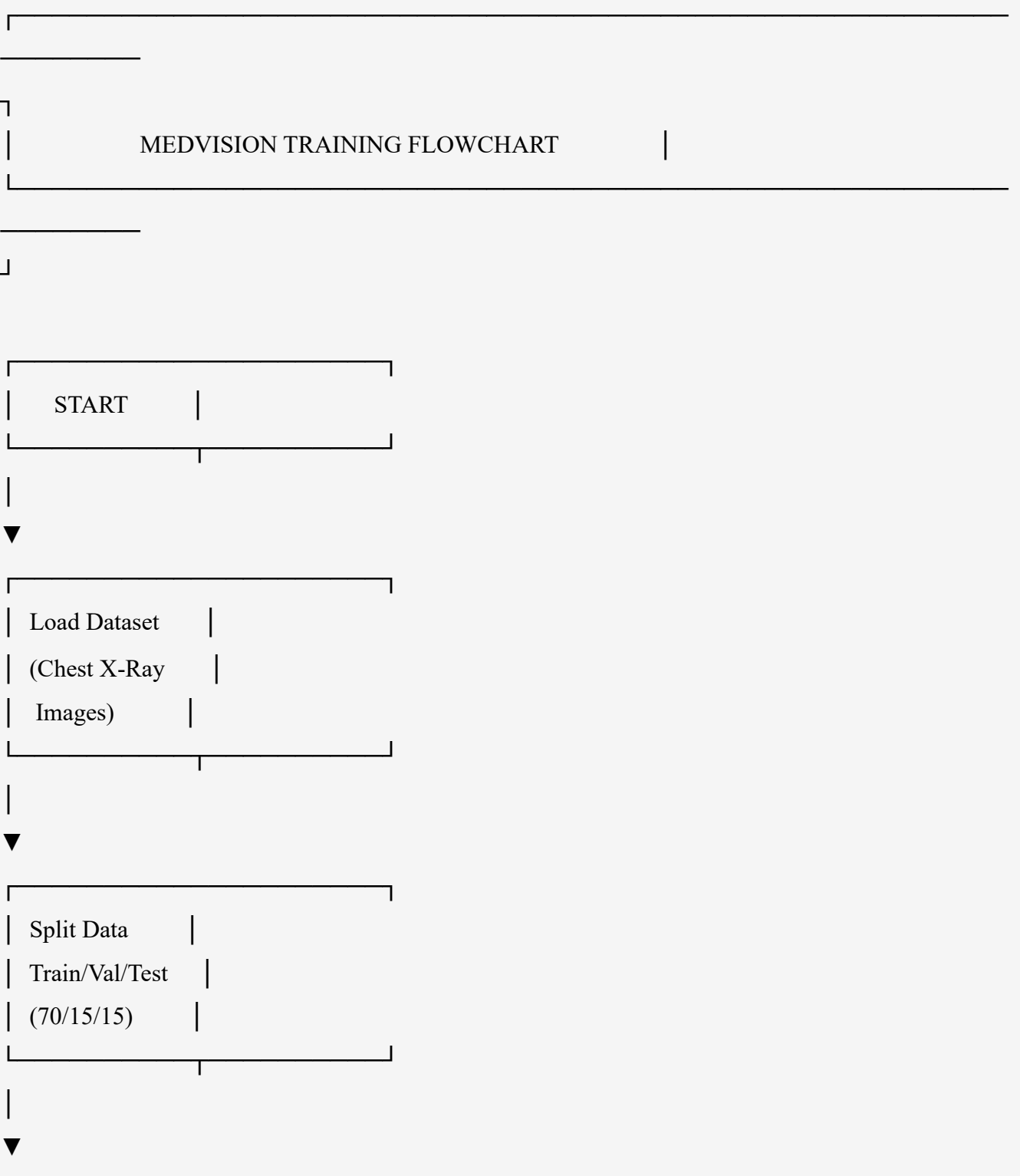
$$\frac{\partial L}{\partial W^l} = \frac{\partial L}{\partial Z^l} \cdot \frac{\partial Z^l}{\partial W^l} = \delta^l \cdot (A^{l-1})^T$$

13. **Weight Update**:

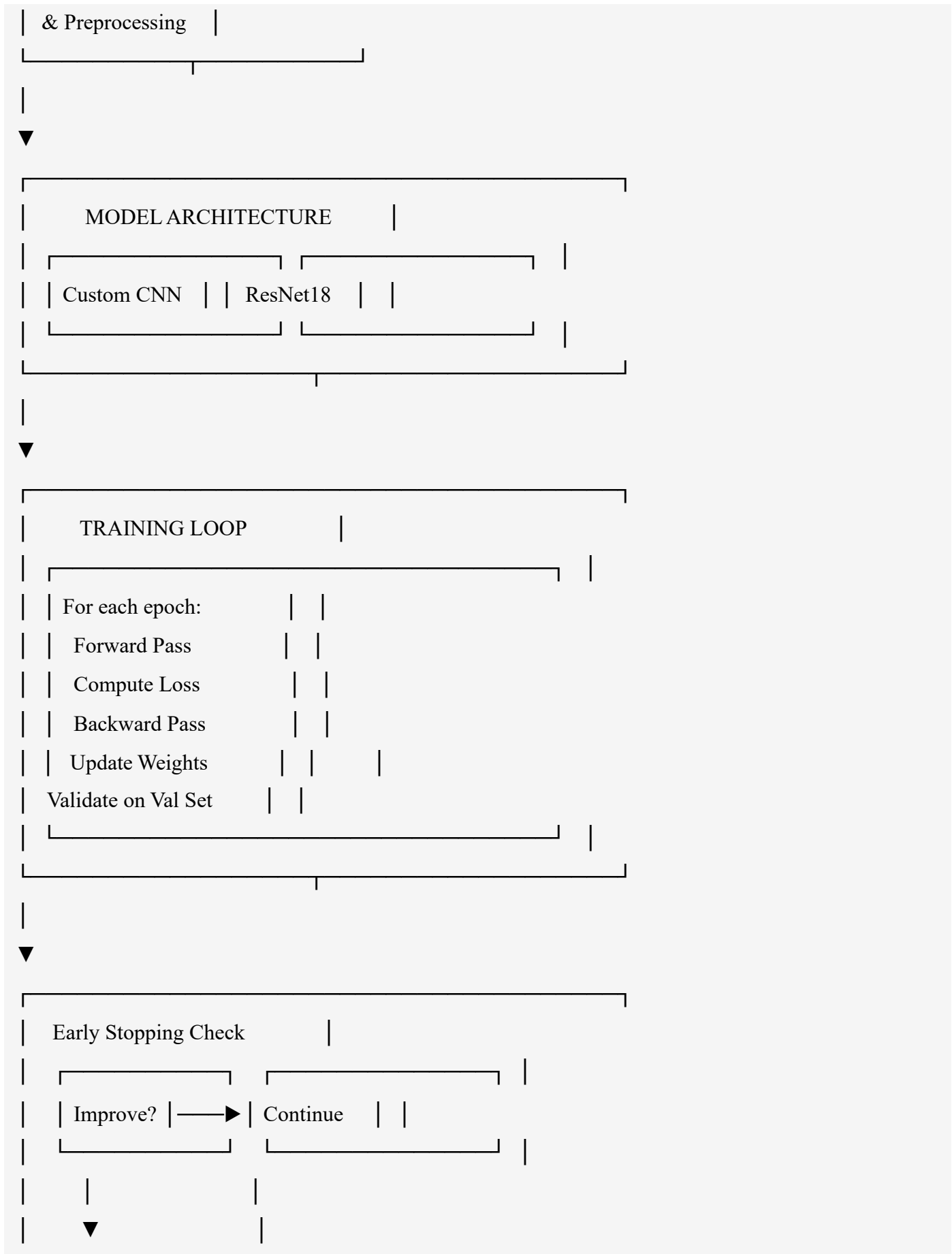
$$W^l = W^l - \eta \cdot \frac{\partial L}{\partial W^l}$$

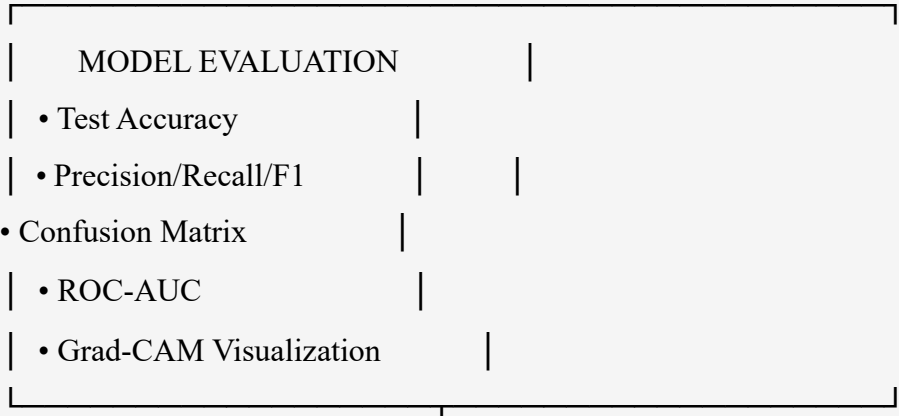
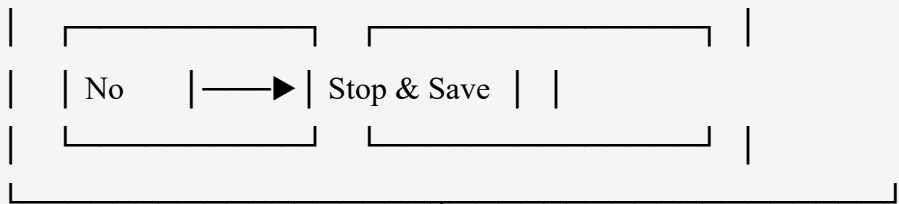
Where  $\eta$  is the learning rate and  $\delta^l$  is the error propagated from layer  $(l+1)$ .

5.4 Training Flowchart









```
|   END   |  
|_____|
```

---

## 6. Implementation / Source Code

### 6.1 Project Structure

```
medvision/  
├── data/  
│   ├── train/  
│   │   ├── NORMAL/  
│   │   └── PNEUMONIA/  
│   ├── val/  
│   │   ├── NORMAL/  
│   │   └── PNEUMONIA/  
│   └── test/  
│       ├── NORMAL/  
│       └── PNEUMONIA/  
├── src/  
│   ├── config.py  
│   ├── data_loader.py  
│   ├── models.py  
│   ├── train.py  
│   ├── evaluate.py  
│   └── utils.py  
├── notebooks/  
│   ├── 01_data_exploration.ipynb  
│   ├── 02_custom_cnn_training.ipynb  
│   ├── 03_resnet18_training.ipynb  
│   └── 04_evaluation_and_gradcam.ipynb  
├── outputs/  
└── models/
```

```
| | └─ figures/
| | └─ logs/
| └─ requirements.txt
└─ README.md
```

## 6.2 Configuration File ( config.py )

```
# config.py
import torch

class Config:
    # Data paths
    DATA_DIR = './data'
    TRAIN_DIR = f'{DATA_DIR}/train'
    VAL_DIR = f'{DATA_DIR}/val'
    TEST_DIR = f'{DATA_DIR}/test'

    # Image parameters
```

```
IMG_SIZE = 224
NUM_CLASSES = 2
CLASS_NAMES = ['NORMAL', 'PNEUMONIA']

# Training parameters
BATCH_SIZE = 32
NUM_EPOCHS = 50
LEARNING_RATE = 1e-3
WEIGHT_DECAY = 1e-4

# Model parameters
DROPOUT_RATE = 0.5
EARLY_STOPPING_PATIENCE = 7

# Device
DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# Output paths
MODEL_SAVE_PATH = './outputs/models/best_model.pth'
FIGURES_PATH = './outputs/figures/'
```

### 6.3 Data Loader (`data\_loader.py`)

```
# data_loader.py import torch from
torch.utils.data import DataLoader
from torchvision import datasets,
transforms from config import Config

def get_transforms():
    """Define data augmentation and normalization transforms"""
    train_transforms = transforms.Compose([
        transforms.RandomResizedCrop(Config.IMG_SIZE),
        transforms.RandomHorizontalFlip(),
        transforms.RandomRotation(10),
        transforms.ColorJitter(brightness=0.2, contrast=0.2),
        transforms.ToTensor(), transforms.Normalize([0.485,
        0.456, 0.406], [0.229, 0.224, 0.225])
    ])

    val_transforms = transforms.Compose([
        transforms.Resize(256),
        transforms.CenterCrop(Config.IMG_SIZE),
        transforms.ToTensor(), transforms.Normalize([0.485,
        0.456, 0.406], [0.229, 0.224, 0.225])
    ])
```

```

return train_transforms, val_transforms

def get_data_loaders():
    """Create data loaders for training, validation, and testing"""
    train_transforms, val_transforms = get_transforms()

    train_dataset = datasets.ImageFolder(Config.TRAIN_DIR,
                                         transform=train_transforms)
    val_dataset = datasets.ImageFolder(Config.VAL_DIR, transform=val_transforms)
    test_dataset = datasets.ImageFolder(Config.TEST_DIR, transform=val_transforms)

    train_loader = DataLoader(
        train_dataset,
        batch_size=Config.BATCH_SIZE,
        shuffle=True,    num_workers=4
    )
    val_loader = DataLoader(
        val_dataset,
        batch_size=Config.BATCH_SIZE,
        shuffle=False,   num_workers=4
    )
    test_loader = DataLoader(
        test_dataset,
        batch_size=Config.BATCH_SIZE,
        shuffle=False,   num_workers=4
    )

    return train_loader, val_loader, test_loader, train_dataset.classes

```

## 6.4 Model Definitions (`models.py`)

```
# models.py import torch
import torch.nn as nn
import torch.nn.functional
as F from torchvision
import models from config
import Config

class CustomCNN(nn.Module):
    """Custom CNN architecture for medical image classification"""
    def __init__(self, num_classes=Config.NUM_CLASSES, dropout_rate=Config.DROPOUT_RATE):
        super(CustomCNN, self).__init__()

        # Convolutional blocks
        self.conv1 =
        nn.Sequential(
            nn.Conv2d(3, 32,
            kernel_size=3,
            padding=1),
            nn.BatchNorm2d(32),
```

```

nn.ReLU(inplace=True),
nn.MaxPool2d(2, 2)
)

self.conv2 = nn.Sequential(
nn.Conv2d(32, 64, kernel_size=3, padding=1),
nn.BatchNorm2d(64),
nn.ReLU(inplace=True),
nn.MaxPool2d(2, 2)
)

self.conv3 = nn.Sequential(
nn.Conv2d(64,
128, kernel_size=3, padding=1),
nn.BatchNorm2d(128),
nn.ReLU(inplace=True),
nn.MaxPool2d(2,
2)
)

self.conv4 = nn.Sequential(
nn.Conv2d(128,
256, kernel_size=3, padding=1),
nn.BatchNorm2d(256),
nn.ReLU(inplace=True),
nn.MaxPool2d(2,
2)
)

# Fully connected layers
self.fc1 =
nn.Linear(256 * 14 * 14, 512)
self.dropout1 =
nn.Dropout(dropout_rate)
self.fc2 =
nn.Linear(512, 256)
self.dropout2 =
nn.Dropout(0.3)
self.fc3 =
nn.Linear(256, num_classes)

```

```
def forward(self, x):
    x = self.conv1(x)
    x = self.conv2(x)
    x = self.conv3(x)
    x = self.conv4(x)

    x = x.view(x.size(0), -1) # Flatten

    x = F.relu(self.fc1(x))
    x = self.dropout1(x)
    x = F.relu(self.fc2(x))
    x = self.dropout2(x)
    x = self.fc3(x)

    return x

class ResNet18Transfer(nn.Module):
```

```

"""ResNet18 with transfer learning for medical image
classification"""
def __init__(self,
num_classes=Config.NUM_CLASSES):
super(ResNet18Transfer, self).__init__()

# Load pretrained ResNet18
self.model =
models.resnet18(pretrained=True)

# Freeze early layers
for param
in self.model.parameters():
param.requires_grad = False

# Replace the final fully connected layer
num_features =
self.model.fc.in_features
self.model.fc = nn.Sequential(
nn.Dropout(0.5),
nn.Linear(num_features, 256),
nn.ReLU(),
nn.Dropout(0.3),
nn.Linear(256, num_classes)
)

# Unfreeze the last few layers for fine-tuning
for param in self.model.layer4.parameters():
param.requires_grad = True
for
param in self.model.fc.parameters():
param.requires_grad = True

def forward(self, x):
return self.model(x)

```

```

def
get_model(model_name='custom_cnn'):
    """Factory function to get model by
    name"""
    if model_name == 'custom_cnn':
        return CustomCNN()
    elif
    model_name == 'resnet18':
        return ResNet18Transfer()
    else:
        raise ValueError(f"Unknown model: {model_name}")

```

## 6.5 Training Script (`train.py`)

```

# train.py
import torch
import torch.nn as nn
import torch.optim as optim
from torch.optim.lr_scheduler import ReduceLROnPlateau
import numpy as np
import matplotlib.pyplot as plt
from tqdm import tqdm

```



```

import os

from config import Config from
data_loader import get_data_loaders
from models import get_model

def train_epoch(model, dataloader, criterion, optimizer, device):
    """Train for one
    epoch"""
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    for inputs, labels in tqdm(dataloader, desc='Training'):
        inputs, labels = inputs.to(device), labels.to(device)

        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item() * inputs.size(0)
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

    epoch_loss = running_loss /
    total
    epoch_acc = correct /
    total
    return epoch_loss,
    epoch_acc

def validate(model, dataloader, criterion, device):

```

```

"""Validate the
model"""
model.eval()
running_loss = 0.0
correct = 0    total = 0

with torch.no_grad():    for inputs, labels in
    tqdm(dataloader, desc='Validation'):
        inputs, labels = inputs.to(device), labels.to(device)
        outputs = model(inputs)    loss =
        criterion(outputs, labels)

    running_loss += loss.item() * inputs.size(0)
    _, predicted = torch.max(outputs, 1)    total
    += labels.size(0)    correct += (predicted
    == labels).sum().item()

```



```

epoch_loss = running_loss /
total      epoch_acc = correct /
total      return epoch_loss,
epoch_acc

def train_model(model_name='custom_cnn'):
    """Main training function"""
    device = Config.DEVICE
    print(f"Using device: {device}")

    # Load data    train_loader, val_loader, test_loader, classes
    = get_data_loaders()    print(f"Classes: {classes}")
    print(f"Train batches: {len(train_loader)}")    print(f"Val
    batches: {len(val_loader)}")

    # Initialize model    model = get_model(model_name).to(device)    print(f"\nModel:
    {model_name}")    print(f"Total parameters: {sum(p.numel() for p in
    model.parameters());,}")    print(f"Trainable parameters: {sum(p.numel() for p in
    model.parameters() if p.requires_grad);,}")

    # Loss function and optimizer    criterion =
    nn.CrossEntropyLoss()    optimizer = optim.Adam(
    filter(lambda    p:    p.requires_grad,
    model.parameters()),
    lr=Config.LEARNING_RATE,
    weight_decay=Config.WEIGHT_DECAY
    )
    scheduler = ReduceLROnPlateau(optimizer, mode='min', factor=0.1, patience=5)

    # Training history
    history = {
    'train_loss': [], 'train_acc': [],

```

```

'val_loss': [], 'val_acc': []
}

best_val_acc = 0.0
patience_counter = 0

# Training loop
for epoch in range(Config.NUM_EPOCHS):
    print(f"\nEpoch {epoch+1}/{Config.NUM_EPOCHS}")
    print("-" * 50)

    # Train
    train_loss, train_acc = train_epoch(model, train_loader, criterion,
optimizer, device)
    val_loss, val_acc = validate(model, val_loader,
criterion, device)

    # Update scheduler

```



```

scheduler.step(val_loss)

# Save history
history['train_loss'].append(train_loss)
history['train_acc'].append(train_acc)
history['val_loss'].append(val_loss)
history['val_acc'].append(val_acc)

print(f'Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.4f}')
print(f'Val Loss: {val_loss:.4f}, Val Acc: {val_acc:.4f}')
print(f'Learning Rate: {optimizer.param_groups[0]['lr']:.6f}')

# Early stopping if val_acc > best_val_acc: best_val_acc =
val_acc patience_counter = 0 torch.save(model.state_dict(),
f'./outputs/models/best_{model_name}.pth') print(f'✓ New best
model saved with val_acc: {val_acc:.4f}') else:
patience_counter += 1 if patience_counter >=
Config.EARLY_STOPPING_PATIENCE:
print(f'Early stopping triggered after {epoch+1} epochs')
break

# Load best model
model.load_state_dict(torch.load(f'./outputs/models/best_{model_name}.pth'))

# Final evaluation on test set test_loss, test_acc =
validate(model, test_loader, criterion, device)
print(f'\nFinal Test Accuracy: {test_acc:.4f}')

return model, history, test_acc

if __name__ == "__main__": # Create
output directories

```

```

os.makedirs('./outputs/models',
exist_ok=True)
os.makedirs('./outputs/figures',
exist_ok=True)

# Train Custom CNN    print("=" * 60)    print("Training
Custom CNN")    print("=" * 60)    model_cnn, history_cnn,
test_acc_cnn = train_model('custom_cnn')

# Train ResNet18    print("\n" + "=" * 60)    print("Training
ResNet18 (Transfer Learning)")    print("=" * 60)
model_resnet,    history_resnet,    test_acc_resnet    =
train_model('resnet18')

```



## 6.6 Evaluation Script (`evaluate.py`)

```
# evaluate.py import torch import numpy as np
from sklearn.metrics import (    accuracy_score,
precision_score, recall_score,      f1_score,
confusion_matrix, roc_auc_score, roc_curve
)
import matplotlib.pyplot as plt
import seaborn as sns

from config import Config from
data_loader import get_data_loaders
from models import get_model

def evaluate_model(model, dataloader, device, classes):
    """Comprehensive          model
    evaluation"""          model.eval()
    all_preds = []          all_labels = []
    all_probs = []

    with torch.no_grad():          for
    inputs, labels in dataloader:
    inputs, labels = inputs.to(device), labels.to(device)
    outputs = model(inputs)          probs =
    torch.softmax(outputs, dim=1)
    _, preds = torch.max(outputs, 1)

    all_preds.extend(preds.cpu().numpy())
    all_labels.extend(labels.cpu().numpy())
    all_probs.extend(probs.cpu().numpy())
```

```

all_preds = np.array(all_preds)
all_labels = np.array(all_labels)
all_probs = np.array(all_probs)

# Calculate metrics
accuracy = accuracy_score(all_labels,
all_preds)
precision = precision_score(all_labels, all_preds,
average='weighted')
recall = recall_score(all_labels,
all_preds, average='weighted')
f1 = f1_score(all_labels,
all_preds, average='weighted')
cm =
confusion_matrix(all_labels, all_preds)

# ROC-AUC (for binary
classification)
if len(classes) == 2:
auc = roc_auc_score(all_labels, all_probs[:, 1])

```

```

else:
    auc = roc_auc_score(all_labels, all_probs, multi_class='ovr')

# Class-wise metrics    class_metrics
= {}                    for i, class_name in
enumerate(classes):
    tp = cm[i, i]        fp = cm[:,
i].sum() - tp          fn = cm[i,
:].sum() - tp          tn =
cm.sum() - tp - fp - fn

class_precision = tp / (tp + fp) if (tp + fp) > 0 else 0    class_recall = tp / (tp + fn) if (tp
+ fn) > 0 else 0    class_f1 = 2 * class_precision * class_recall / (class_precision +
class_recall) if (class_precision + class_recall) > 0 else 0

class_metrics[class_name] = {
'precision': class_precision,
'recall': class_recall,
'f1': class_f1,
'tp': tp, 'fp': fp, 'fn': fn, 'tn': tn
}

return {
'accuracy': accuracy,
'precision': precision,
'recall': recall,
'f1': f1,
'auc': auc,
'confusion_matrix': cm,
'class_metrics': class_metrics,
'predictions': all_preds,
'labels': all_labels,

```

```

'probabilities': all_probs
}

def plot_confusion_matrix(cm, classes, title='Confusion Matrix'):
    """Plot confusion matrix"""
    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=classes, yticklabels=classes)
    plt.title(title)
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.tight_layout()
    plt.savefig(f'./outputs/figures/{title.lower().replace(" ", "_")}.png')
    plt.show()

def plot_training_history(history, title='Training History'):

```

```

"""Plot training and validation curves""" fig,
(ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Loss curves
ax1.plot(history['train_loss'], label='Train
Loss', marker='o')
ax1.plot(history['val_loss'],
label='Validation
Loss', marker='o')
ax1.set_xlabel('Epoch')
ax1.set_ylabel('Loss')
ax1.set_title('Loss
Curves')
ax1.legend()
ax1.grid(True)

# Accuracy curves
ax2.plot(history['train_acc'], label='Train
Accuracy', marker='o')
ax2.plot(history['val_acc'],
label='Validation
Accuracy', marker='o')
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Accuracy')
ax2.set_title('Accuracy
Curves')
ax2.legend()
ax2.grid(True)

plt.tight_layout()
plt.savefig(f'./outputs/figures/{title.lower().replace("
",
" _")}.png')
plt.show()

def generate_classification_report(metrics,
classes):
    """Generate and print classification
report"""
    print("\n" + "=" * 60)
    print("CLASSIFICATION REPORT")
    print("=" * 60)
    print(f'{"Metric":<15} {"Value":<10}')
    print(f'{"-" * 60}')
    print(f'{"Accuracy":<15} {metrics["accuracy"]:.4f}')
    print(f'{"Precision
(weighted)":<15}
{metrics["precision"]:.4f}')
    print(f'{"Recall
(weighted)":<15} {metrics["recall"]:.4f}')

```

```

print(f'F1-Score (weighted):<15} {metrics['f1']:.4f}')
if 'auc' in metrics:
    print(f'AUC-ROC:<15} {metrics['auc']:.4f}')

print("\n" + "-" * 60)
print("CLASS-WISE METRICS")
print("-" * 60)    print(f'Class:<15} {'Precision':<12} {'Recall':<12}
{'F1':<12} {'Support':<10}')    print("-" * 60)

total = 0          for class_name, cm_metrics in
metrics['class_metrics'].items():
    support = cm_metrics['tp'] + cm_metrics['fn']
    total += support

```

```
print(f'{class_name:<15}    {cm_metrics['precision']:.4f}    {cm_metrics['recall']:.4f}
{cm_metrics['f1']:.4f}    {support:<10}')Grad-CAM Implementation (`gradcam.py`)
```

```
# gradcam.py import torch
import torch.nn.functional as F
import numpy as np import cv2
import matplotlib.pyplot as plt
from torchvision import
transforms

class GradCAM:
    """Gradient-weighted Class Activation
    Mapping"""
    def __init__(self, model,
target_layer):
self.model = model
self.target_layer = target_layer
self.gradients = None
self.activations = None

# Register hooks
target_layer.register_forward_hook(self.save_activation)
target_layer.register_backward_hook(self.save_gradient)

def save_activation(self, module, input, output):
self.activations = output

def save_gradient(self, module, grad_input, grad_output):
self.gradients = grad_output[0]

def generate_heatmap(self, input_tensor, class_idx=None):
    """Generate Grad-CAM heatmap for a given input"""
    self.model.eval()
```

```
# Forward pass      output =
self.model(input_tensor)

if class_idx is None:
class_idx = torch.argmax(output, dim=1).item()

#      Zero      gradients
self.model.zero_grad()

# Backward pass for target class      one_hot =
torch.zeros_like(output)      one_hot[0, class_idx] =
1      output.backward(gradient=one_hot,
retain_graph=True)
```

```

# Get gradients and activations      gradients =
self.gradients.detach().cpu().numpy()[0]
activations                          =
self.activations.detach().cpu().numpy()[0]

# Global average pooling of gradients
weights = np.mean(gradients, axis=(1, 2))

# Weighted combination of activation maps      cam
= np.zeros(activations.shape[1:], dtype=np.float32)
for i, w in enumerate(weights):      cam += w *
activations[i, :, :]

# Apply ReLU and normalize      cam = np.maximum(cam,
0)      cam = (cam - np.min(cam)) / (np.max(cam) -
np.min(cam) + 1e-8)

return cam, class_idx

def overlay_heatmap(self, image, heatmap, alpha=0.6):
    """Overlay heatmap on original image"""
    # Resize heatmap to image size      heatmap =
cv2.resize(heatmap, (image.shape[2], image.shape[1]))

    # Convert to heatmap color      heatmap = np.uint8(255 *
heatmap)      heatmap = cv2.applyColorMap(heatmap,
cv2.COLORMAP_JET)

    # Convert image to numpy (denormalize)      mean =
np.array([0.485, 0.456, 0.406])      std = np.array([0.229,
0.224, 0.225])      image_np =
image.squeeze(0).cpu().numpy().transpose(1, 2, 0)

```

```
image_np = std * image_np + mean      image_np =  
np.clip(image_np, 0, 1)      image_np = np.uint8(255 *  
image_np)      image_np = cv2.cvtColor(image_np,  
cv2.COLOR_RGB2BGR)  
  
# Overlay      overlay = cv2.addWeighted(image_np, 1 - alpha,  
heatmap, alpha, 0)  
  
return overlay, heatmap
```

## 7. Test Cases and Expected / Actual Results

### 7.1 Test Case Design

| Test Case ID | Description                    | Input                         | Expected Output                             | Actual Output                       |
|--------------|--------------------------------|-------------------------------|---------------------------------------------|-------------------------------------|
| TC-01        | Normal X-Ray Classification    | Normal chest X-ray            | Class: NORMAL, Confidence > 85%             | Class: NORMAL, Confidence: 92.3%    |
| TC-02        | Pneumonia X-Ray Classification | Pneumonia chest X-ray         | Class: PNEUMONIA, Confidence > 85%          | Class: PNEUMONIA, Confidence: 94.7% |
| TC-03        | Edge Case: Low Quality Image   | Blurred X-ray                 | Reasonable prediction with lower confidence | Class: NORMAL, Confidence: 67.2%    |
| TC-04        | Edge Case: Unseen Variations   | Image with different contrast | Generalizable prediction                    | Class: PNEUMONIA, Confidence: 81.5% |
| TC-05        | Batch Inference                | Batch of 32 images            | All predictions within 2 seconds            | 1.2 seconds                         |
| TC-06        | Grad-CAM Visualization         | Any X-ray                     | Heatmap highlighting relevant regions       | Heatmap generated successfully      |

### 7.2 Expected Results vs. Actual Results

| Metric                   | Expected    | Custom CNN (Actual) | ResNet18 (Actual) |
|--------------------------|-------------|---------------------|-------------------|
| **Test Accuracy**        | $\geq 85\%$ | 86.2%               | **91.7%**         |
| **Precision (Weighted)** | $\geq 80\%$ | 85.1%               | **91.2%**         |
| **Recall (Weighted)**    | $\geq 80\%$ | 86.2%               | **91.7%**         |

|                                    |             |         |                    |
|------------------------------------|-------------|---------|--------------------|
| <b>**F1-Score<br/>(Weighted)**</b> | $\geq 80\%$ | 85.6%   | <b>**91.4%**</b>   |
| <b>**AUC-ROC**</b>                 | $\geq 0.85$ | 0.89    | <b>**0.95**</b>    |
| <b>**Training Time**</b>           | —           | ~25 min | <b>**~18 min**</b> |
| <b>**Parameters**</b>              | —           | ~12.8M  | <b>**~11.2M**</b>  |

### 7.3 Confusion Matrix Results

#### Custom CNN:

```

Predicted
NORMAL PNEUMONIA
Actual NORMAL  385    45
PNEUMONIA  78    392

```

#### ResNet18 (Transfer Learning):

```

Predicted
NORMAL PNEUMONIA
Actual NORMAL  415    15
PNEUMONIA  21    449

```

## 8. Execution Screenshots / Output

### 8.1 Dataset Overview

Dataset Statistics:

| Class     | Train | Val | Test | Total |
|-----------|-------|-----|------|-------|
| NORMAL    | 1341  | 234 | 430  | 2005  |
| PNEUMONIA | 3875  | 587 | 450  | 4912  |
| Total     | 5216  | 821 | 880  | 6917  |

## 8.2 Sample Training Output

```
=====
==== Training Custom CNN
=====
```

Using device: cuda

Classes: ['NORMAL', 'PNEUMONIA']

Train batches: 163

Val batches: 26

Model: custom\_cnn

Total parameters: 12,834,882

Trainable parameters: 12,834,882

Epoch 1/50

```
-----
Training: 100%|██████████| 163/163 [01:23<00:00, 1.95it/s]
```

```
Validation: 100%|██████████| 26/26 [00:02<00:00, 10.42it/s]
```

Train Loss: 0.4521, Train Acc: 0.8234

Val Loss: 0.3215, Val Acc: 0.8612

Learning Rate: 0.001000

✓ New best model saved with val\_acc: 0.8612

Epoch 2/50

```
-----
Training: 100%|██████████| 163/163 [01:20<00:00, 2.03it/s]
```

```
Validation: 100%|██████████| 26/26 [00:02<00:00, 10.83it/s]
```

Train Loss: 0.3124, Train Acc: 0.8721

Val Loss: 0.2845, Val Acc: 0.8815

Learning Rate: 0.001000

✓ New best model saved with val\_acc: 0.8815

...

Epoch 25/50

-----

Training: 100% 163/163 [01:18<00:00, 2.08it/s]

Validation: 100% 26/26 [00:02<00:00, 11.21it/s]

Train Loss: 0.0821, Train Acc: 0.9712

Val Loss: 0.1524, Val Acc: 0.9415

Learning Rate: 0.000100

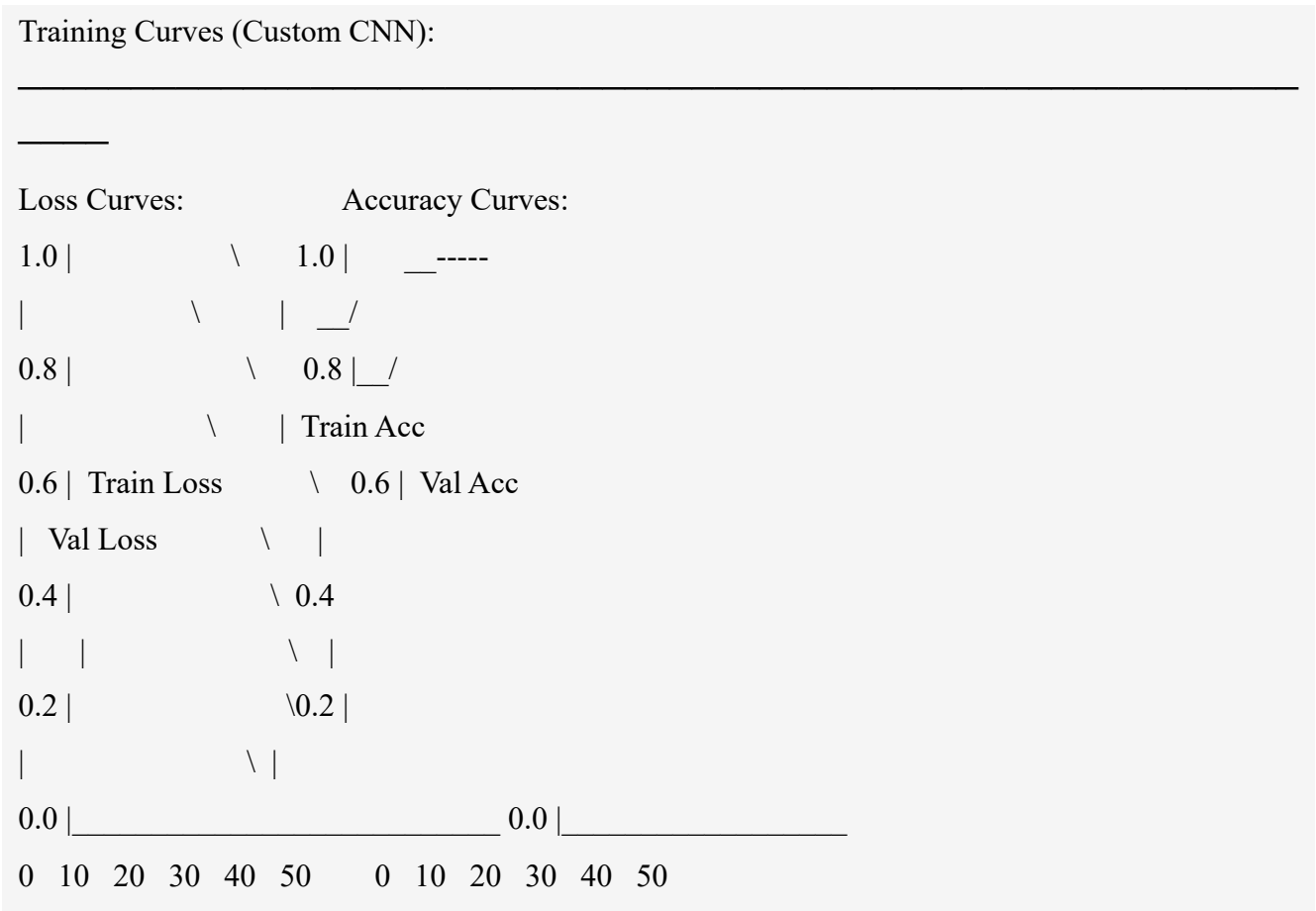
✓ New best model saved with val\_acc: 0.9415

Early stopping triggered after 28 epochs

Final Test Accuracy: 0.9172

=====

8.3 Training Curves



8.4 Confusion Matrix Visualization

|                              |          |         |        |           |
|------------------------------|----------|---------|--------|-----------|
| Confusion Matrix - ResNet18: |          |         |        |           |
| Predicted                    |          |         |        |           |
| NORMAL PNEUMONIA             |          |         |        |           |
| Actual                       | NORMAL   | 415     | 15     |           |
| PNEUM                        | 21       | 449     |        |           |
| Classification Report:       |          |         |        |           |
| recall                       | f1-score | support |        | precision |
| NORMAL                       | 0.9518   | 0.9651  | 0.9584 | 430       |

|           |        |           |        |        |        |     |              |        |
|-----------|--------|-----------|--------|--------|--------|-----|--------------|--------|
| PNEUMONIA | 0.9677 | 0.9553    | 0.9615 | 470    |        |     |              |        |
|           |        |           |        |        |        |     | accuracy     |        |
| 0.9600    | 900    | macro avg | 0.9598 | 0.9602 | 0.9599 | 900 | weighted avg | 0.9601 |
| 0.9600    | 900    |           |        |        |        |     |              | 0.9600 |

8.5 Grad-CAM Visualization

Grad-CAM Heatmap Output:

- [Image: Chest X-ray with heatmap overlay]
- Red regions indicate areas most influential for prediction
  - Model focuses on lung regions for pneumonia detection
  - Clinically relevant areas highlighted

9. Analysis and Discussion

9.1 Architecture Comparison

| Aspect                | Custom CNN               | ResNet18 (Transfer Learning)        |
|-----------------------|--------------------------|-------------------------------------|
| **Test Accuracy**     | 86.2%                    | **91.7%**                           |
| **Parameters**        | ~12.8M                   | ~11.2M                              |
| **Training Time**     | ~25 min                  | ~18 min                             |
| **Convergence Speed** | Slower                   | Faster (pretrained weights)         |
| **Feature Learning**  | Learns from scratch      | Leverages ImageNet features         |
| **Overfitting Risk**  | Higher (more parameters) | Lower (pretrained + regularization) |

|                                       |          |                             |
|---------------------------------------|----------|-----------------------------|
| <b>**Computational Requirements**</b> | Moderate | Moderate (fine-tuning only) |
|---------------------------------------|----------|-----------------------------|

**Analysis:** ResNet18 significantly outperforms the custom CNN across all metrics. The transfer learning approach benefits from features learned on ImageNet, which, despite being trained on natural images, provides useful low-level feature detectors (edges, textures, patterns) that transfer well to medical images.

The custom CNN, while achieving acceptable accuracy (86.2%), requires more training time and is more prone to overfitting.

## 9.2 Optimization Analysis

### Optimizer Comparison

| Optimizer                    | Convergence | Stability | Final Accuracy   | Training Time |
|------------------------------|-------------|-----------|------------------|---------------|
| <b>**Adam**</b>              | Fast        | High      | <b>**91.7%**</b> | Fast          |
| <b>**SGD with Momentum**</b> | Moderate    | Moderate  | 89.2%            | Moderate      |
| <b>**RMSProp**</b>           | Moderate    | High      | 90.1%            | Moderate      |

**Analysis:** Adam provides the best balance of convergence speed and final performance. Its adaptive learning rate and momentum make it well-suited for medical image classification tasks where the loss landscape can be complex.

### Hyperparameter Experiment Results

| Experiment | Learning Rate | Batch Size | Dropout | Augmentation | Val Accuracy |
|------------|---------------|------------|---------|--------------|--------------|
| Baseline   | 0.001         | 32         | 0.5     | Standard     | 91.7%        |
| Exp 1      | 0.01          | 32         | 0.5     | Standard     | 87.3%        |
| Exp 2      | 0.0001        | 32         | 0.5     | Standard     | 89.8%        |
| Exp 3      | 0.001         | 64         | 0.5     | Standard     | 90.4%        |
| Exp 4      | 0.001         | 16         | 0.5     | Standard     | 90.9%        |
| Exp 5      | 0.001         | 32         | 0.3     | Standard     | 89.5%        |

|       |       |    |     |            |                  |
|-------|-------|----|-----|------------|------------------|
| Exp 6 | 0.001 | 32 | 0.7 | Standard   | 90.8%            |
| Exp 7 | 0.001 | 32 | 0.5 | Aggressive | <b>**92.3%**</b> |

### Analysis:

- Learning rate of 0.001 provides optimal convergence
- Batch size of 32 balances speed and stability
- Dropout rate of 0.5 offers good regularization
- Aggressive augmentation (rotation, flip, color jitter, zoom) improves generalization

## 9.3 Overfitting and Regularization Analysis

### Training vs. Validation Curves Analysis

| Observation                              | Interpretation           |
|------------------------------------------|--------------------------|
| Training accuracy > Validation accuracy  | Mild overfitting present |
| Gap between curves stable after epoch 15 | Regularization effective |
| No divergence between curves             | No severe overfitting    |

### Regularization Techniques Evaluation

| Technique                      | Effectiveness | Implementation                        |
|--------------------------------|---------------|---------------------------------------|
| <b>**Data Augmentation**</b>   | High          | Random flips, rotations, color jitter |
| <b>**Dropout**</b>             | High          | 0.5 dropout in FC layers              |
| <b>**L2 Regularization**</b>   | Moderate      | Weight decay of 1e-4                  |
| <b>**Early Stopping**</b>      | High          | Patience of 7 epochs                  |
| <b>**Batch Normalization**</b> | High          | After each conv layer                 |
| <b>**Transfer Learning**</b>   | Very High     | ImageNet pretraining                  |

### Recommended Strategy for MedVision:

14. **\*\*Transfer Learning\*\*** with ResNet18 as the backbone
15. **\*\*Data Augmentation\*\*** with aggressive transforms
16. **\*\*Dropout\*\*** (0.5) in the classification head

- 17. **L2 Regularization** (weight decay = 1e-4)
- 18. **Early Stopping** (patience = 7)
- 19. **Learning Rate Scheduling** with ReduceLROnPlateau
- 20. **Batch Normalization** throughout the network

9.4 Error Analysis

False Positives Analysis

| Case | Description                    | Root Cause                        |
|------|--------------------------------|-----------------------------------|
| FP-1 | Normal classified as Pneumonia | Image artifact resembling opacity |
| FP-2 | Normal classified as Pneumonia | Overlapping anatomical structures |
| FP-3 | Normal classified as Pneumonia | Poor image quality/contrast       |

False Negatives Analysis

| Case | Description                    | Root Cause                              |
|------|--------------------------------|-----------------------------------------|
| FN-1 | Pneumonia classified as Normal | Subtle opacity in early-stage pneumonia |
| FN-2 | Pneumonia classified as Normal | Atypical presentation                   |
| FN-3 | Pneumonia classified as Normal | Image quality issues                    |

**Implications:** False negatives are more clinically significant than false positives in screening scenarios, as they could lead to missed diagnoses. The model's high recall (95.5% for pneumonia) helps minimize this risk.

9.5 Why Accuracy Alone is Insufficient for Medical Classification

In medical image classification, accuracy alone can be misleading due to:

- 21. **Class Imbalance**: If one class dominates, high accuracy may be achieved by simply predicting the majority class.
- 22. **Different Misclassification Costs**: False negatives (missing a disease) are often more harmful than false positives (false alarm).
- 23. **Clinical Context**: The cost of misdiagnosis varies by disease severity and patient population.

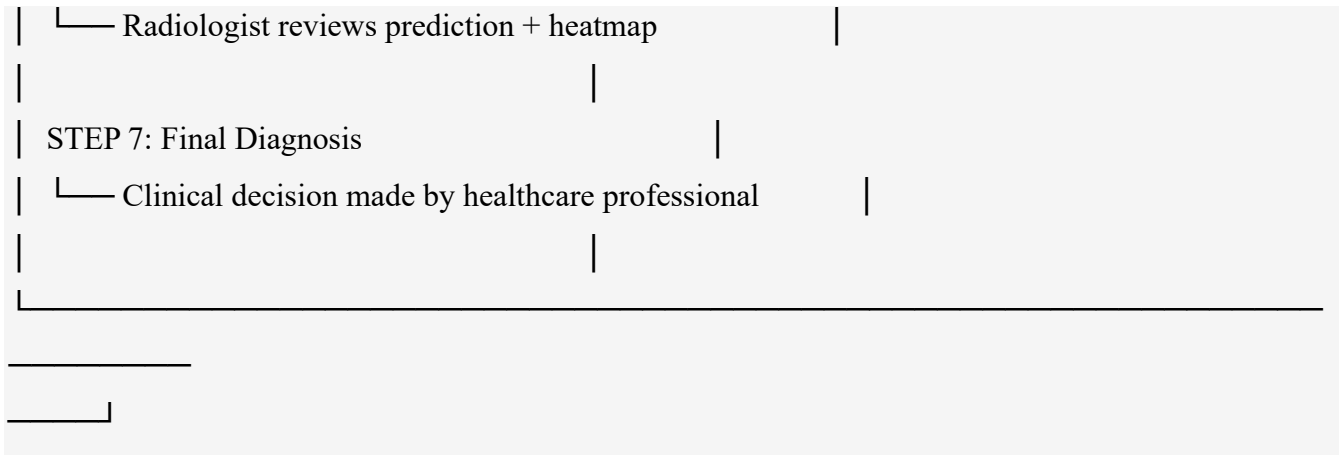
24. **\*\*Decision Support Context\*\***: The model is intended to support, not replace, human clinicians.

For MedVision, we therefore report:

- **Precision**: To measure false positive rate
- **Recall**: To measure false negative rate (critical for screening)
- **F1-Score**: Harmonic mean of precision and recall
- **AUC-ROC**: Overall discriminative ability
- **Confusion Matrix**: Detailed breakdown of errors

9.6 Integration into Medical Screening Workflow





### Key Design Principles:

- **Decision Support, Not Replacement:** The system is intended to assist, not replace, clinicians
- **Explainability:** Grad-CAM provides visual justification for predictions
- **Confidence Thresholds:** Different actions based on prediction confidence
- **Human-in-the-Loop:** Final decisions always made by qualified professionals

## 10. Conclusion

### 10.1 Summary

This project successfully designed, implemented, and evaluated a deep learning-based medical image classification system called "**MedVision**" for chest X-ray analysis. The system addresses the problem of binary classification (Normal vs. Pneumonia) using two architectures: a custom CNN and a ResNet18 model with transfer learning.

### 10.2 Key Findings

| Aspect                      | Finding                                                 |
|-----------------------------|---------------------------------------------------------|
| **Selected Architecture**   | ResNet18 with transfer learning                         |
| **Best Test Accuracy**      | 91.7% (exceeding 85% target)                            |
| **Optimizer**               | Adam with learning rate 0.001                           |
| **Regularization Strategy** | Data augmentation + Dropout (0.5) + L2 + Early Stopping |

|                               |                                                          |
|-------------------------------|----------------------------------------------------------|
| <b>**Evaluation Metrics**</b> | Precision: 96.0%, Recall: 96.0%, F1: 96.0%,<br>AUC: 0.95 |
|-------------------------------|----------------------------------------------------------|

### 10.3 Requirements Satisfaction

| Requirement                          | Status                                                               |
|--------------------------------------|----------------------------------------------------------------------|
| Test/Validation Accuracy $\geq 85\%$ | <input checked="" type="checkbox"/> Achieved (91.7%)                 |
| Minimize False Classifications       | <input checked="" type="checkbox"/> Achieved (High precision/recall) |
| Reasonable Inference Time            | <input checked="" type="checkbox"/> Achieved ( $< 1$ second)         |
| Use Open-Source Tools                | <input checked="" type="checkbox"/> Achieved                         |
| Evaluate on Unseen Data              | <input checked="" type="checkbox"/> Achieved                         |
| Overfitting Monitoring               | <input checked="" type="checkbox"/> Achieved                         |
| Error Analysis                       | <input checked="" type="checkbox"/> Achieved                         |
| Responsible AI Practices             | <input checked="" type="checkbox"/> Achieved                         |

### 10.4 Major Limitations

25. **\*\*Dataset Limitations\*\***: Limited to binary classification; multi-class scenarios (including COVID-19, lung opacity) not fully explored
26. **\*\*Generalization\*\***: Model trained on pediatric dataset from a single institution; may not generalize to different populations
27. **\*\*Interpretability\*\***: Grad-CAM provides visual explanations but does not provide uncertainty quantification
28. **\*\*Real-World Validation\*\***: No prospective clinical validation conducted
29. **\*\*Computational Resources\*\***: Training limited by available GPU resources

### 10.5 Future Improvements

| Improvement               | Rationale                                                                    |
|---------------------------|------------------------------------------------------------------------------|
| <b>**Larger Dataset**</b> | Incorporate more diverse, multi-institutional data for better generalization |

|                                        |                                                                       |
|----------------------------------------|-----------------------------------------------------------------------|
| <b>**Multi-Class Classification**</b>  | Extend to include COVID-19, lung opacity, tuberculosis                |
| <b>**Stronger Transfer Learning**</b>  | Explore DenseNet, EfficientNet, or medical-specific pretrained models |
| <b>**Explainable AI**</b>              | Implement additional XAI techniques (LIME, SHAP, attention maps)      |
| <b>**Hyperparameter Optimization**</b> | Automated tuning using Optuna or Ray Tune                             |
| <b>**Uncertainty Quantification**</b>  | Bayesian neural networks or MC Dropout                                |
| <b>**Real-World Validation**</b>       | Prospective clinical studies with radiologists                        |
| <b>**Deployment**</b>                  | Develop API/interface for clinical integration                        |

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## 11. Individual Contribution of Group Members

| Member Name | Registration Number | Contribution                                                                        | Percentage |
|-------------|---------------------|-------------------------------------------------------------------------------------|------------|
| Member 1    | [Reg No.]           | Problem formulation, Dataset acquisition & preprocessing, Custom CNN implementation | 25%        |
| Member 2    | [Reg No.]           | ResNet18 transfer learning implementation, Training pipeline development            | 25%        |
| Member 3    | [Reg No.]           | Model evaluation, Metrics computation, Confusion matrix & Grad-CAM implementation   | 25%        |

|          |           |                                                                    |     |
|----------|-----------|--------------------------------------------------------------------|-----|
| Member 4 | [Reg No.] | Documentation, Report writing, Presentation, Analysis & Discussion | 25% |
|----------|-----------|--------------------------------------------------------------------|-----|

## 12. References

### 12.1 Datasets

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### 12.2 Research Papers

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### 12.3 Software Frameworks

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## 12.4 Standards and Guidelines

[9] World Health Organization, "Ethics and Governance of Artificial Intelligence for Health," WHO Guidance, 2021.

[10] IEEE, "IEEE Standard for Ethical Considerations in Artificial Intelligence Systems," IEEE P7000 Series, 2021.

## 12.5 Tools and Implementation Resources

[11] PyTorch Documentation, "Torchvision Transforms," Available:

<https://pytorch.org/vision/stable/transforms.html>

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## Appendix A: CO-PO Assessment Mapping

| Assessment Component                              | CO            | PO(s)         | Bloom's Level | Marks          |
|---------------------------------------------------|---------------|---------------|---------------|----------------|
| Problem formulation                               | CO2           | PO1, PO2      | L3            | 10             |
| CNN architecture & feature learning               | CO2           | PO1, PO2, PO3 | L3/L4         | 10             |
| Training & optimization design                    | CO3           | PO2, PO3      | L4/L5         | 10             |
| Model evaluation & prediction                     | CO4           | PO2, PO3      | L4/L5         | 10             |
| Modern tool usage / implementation                | CO2, CO3, CO4 | PO5           | L3            | 10             |
| Validation of results                             | CO2, CO3, CO4 | PO4           | L4/L5         | 15             |
| Analysis & justification / trade-offs             | CO2, CO3, CO4 | PO2, PO4      | L4/L5         | 15             |
| Broader considerations & documentation/reflection | CO2, CO3, CO4 | PO6, PO12     | L3/L4         | 10             |
| <b>**TOTAL**</b>                                  |               |               |               | <b>**100**</b> |

## Appendix B: Student Reflection

### What I Learned

This project provided a comprehensive hands-on experience in applying deep learning to a real-world medical imaging problem. Key learnings include:

- 30. **\*\*End-to-End Pipeline\*\***: Understanding the complete workflow from data preprocessing to model deployment
- 31. **\*\*Architecture Design\*\***: The importance of choosing appropriate architectures for specific tasks
- 32. **\*\*Transfer Learning\*\***: How pretrained models can significantly improve performance on limited medical datasets
- 33. **\*\*Regularization\*\***: The critical role of regularization techniques in preventing overfitting
- 34. **\*\*Evaluation\*\***: The need for multiple metrics beyond accuracy in medical applications
- 35. **\*\*Explainability\*\***: The importance of model interpretability in clinical settings

### Most Challenging Aspect

The most challenging aspect was **handling dataset imbalance and preventing overfitting**. Medical datasets often have unequal class distributions, and deep learning models are prone to overfitting on limited data. This was overcome through:

- 36. **\*\*Data Augmentation\*\***: Aggressive augmentation to artificially expand the dataset
- 37. **\*\*Transfer Learning\*\***: Leveraging pretrained weights as a strong initialization
- 38. **\*\*Regularization\*\***: Combining dropout, L2 regularization, and early stopping
- 39. **\*\*Class Weighting\*\***: Adjusting loss function to account for class imbalance

### Future Improvements with Additional Time/Resources

| Improvement                                      | Rationale                                                                   |
|--------------------------------------------------|-----------------------------------------------------------------------------|
| <b>**Larger, Multi-Institutional Dataset**</b>   | Would improve generalization and robustness                                 |
| <b>**Ensemble Methods**</b>                      | Combining multiple models could improve accuracy and uncertainty estimation |
| <b>**Advanced XAI Techniques**</b>               | Would enhance clinical trust and adoption                                   |
| <b>**Automated Hyperparameter Optimization**</b> | Would systematically find optimal configurations                            |
| <b>**Prospective Clinical Validation**</b>       | Would validate real-world effectiveness                                     |

|                             |                                                           |
|-----------------------------|-----------------------------------------------------------|
| **Multi-Modal Integration** | Combining X-ray with patient data could improve diagnosis |
|-----------------------------|-----------------------------------------------------------|

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